Capital Bikeshare demand study: buy bikes or rebalance?

Ayush Pandey · Round 2 case study submission · November 2026
Data: UCI Bike Sharing Dataset, Washington D.C., hourly rentals for 2011 and 2012

The decision in one table

The CEO wanted to know whether riders' complaints mean the company needs more bikes. I looked at four possible responses and rated each one against what the data shows.

Option	What the data says	Verdict
Buy more bikes	The busiest hour in two years has about 326 bikes on the road at once. With a reserve and broken bikes allowed for, that needs about 453 bikes, against 2,000 owned.	**Not now**
Rebalance using a forecast	Half of a working day's rides fall in six commute hours. The model forecasts hourly demand a month ahead with 17% error.	**Do first, as a pilot**
Repair more bikes faster	The share of broken bikes isn't in the data, but every repaired bike is free capacity.	**Do in parallel**
Collect station data	Without station-level counts nobody can see where bikes actually run out.	**Needed to confirm**

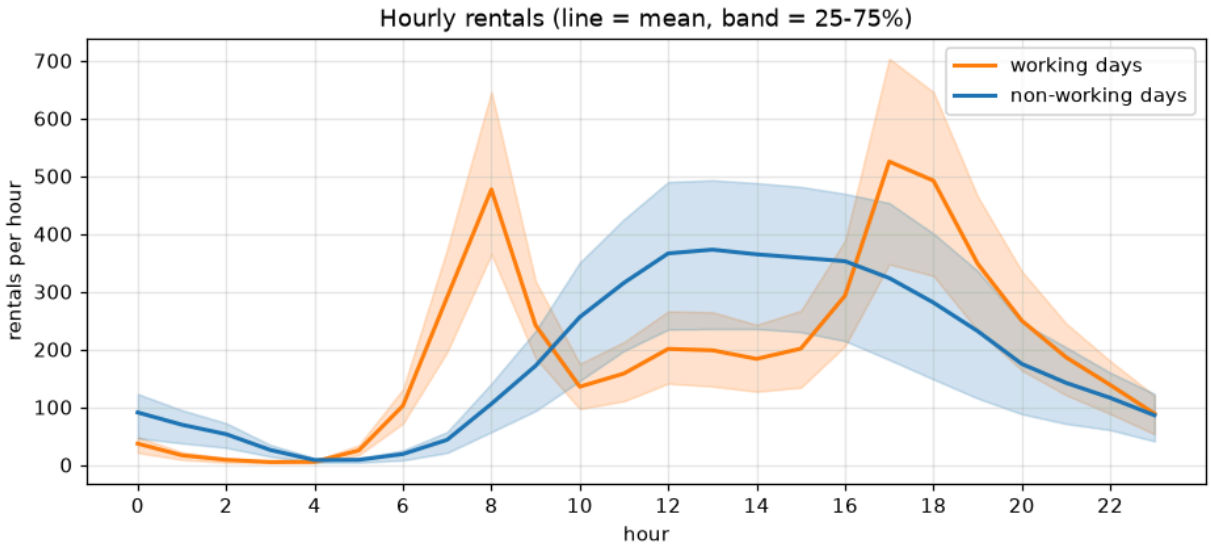
In short, the fleet looks big enough. The bikes are probably in the wrong places at the wrong times. That is a strong lead, not a proof, because the dataset only has citywide totals.

A. Data checks and features

Before any analysis I checked whether the two files (17,379 hourly rows, 731 daily rows) could be trusted. Most checks passed: no blanks or duplicates, casual + registered always equals the total, and the hourly rows add up exactly to the daily file. These were the problems:

Issue found	Details	What I did
Missing hours	165 hours absent. 66 days miss 1–2 night hours, 2 miss the 2 am hour when clocks go forward, 8 have long gaps (e.g. Hurricane Sandy, 29–30 Oct 2012)	Never filled in. The 8 long-gap days are left out of training and out of the main scores
Season labels	Codes follow astronomical seasons (code 1 starts 21 Dec, so it is winter), not the README names	Labelled correctly in charts
Feels-like temperature	0.99 correlated with temperature, and stuck on 17 Aug 2012	Dropped
Humidity of 0%	22 hours on 10 Mar 2011, which is impossible	Kept and noted (one day)
Heavy-rain category	Only 3 hours	Merged with light rain
Casual / registered counts	They add up to the target	Never used as inputs (would leak the answer)

Features used: hour, working day, holiday, weekday, month, season, year, weather type, temperature, humidity and wind. No lag features, because the model is meant for planning a day ahead, when the last few hours' counts aren't known yet.



Hourly rides on working vs non-working days. Lines are averages; the shaded bands cover the middle half of hours.

Three things stood out. Working days have sharp peaks at 8 am and 5–6 pm, driven almost entirely by registered riders, while weekends have one broad midday hump. Rides climb with temperature up to about 30 °C and drop in rain at any temperature. And 2012 had 65% more rides than 2011, with uneven growth by month, which the data cannot explain.

B. Hypothesis test

Claim	Working days put a bigger share of their rides into commute hours (7–10 am, 5–8 pm) than non-working days, even after allowing for month and weather.
Why it matters	If demand arrives in two short bursts, stations can run dry at 8 am even when the city has plenty of bikes overall.
Unit	One value per day (that day's commute share), because hours of the same day aren't independent. 723 days after removing long-gap days.
Method	Compare working and non-working days only within the same month and weather type, then average the differences. p-value from 10,000 shuffles of the working-day label within those groups, plus a bootstrap 95% interval. Threshold 0.05, set beforehand.
Result	51.7% vs 26.0% on average. Within matched groups the gap is **+25.5 points** (95% CI 24.7–26.1), p = 0.0001. None of the 10,000 shuffles came above 4.3 points.
Robustness	+25.2 points on complete days only, +25.6 in 2011, +25.4 in 2012.
Limit	It shows *when* demand peaks, not *where*. That needs trip start and end points.

C. Demand model and number of bikes

How the model was tested

The model predicts rides per hour from the calendar and the weather. To keep the test honest it always predicts a month it hasn't seen, after training on the months before it. I compared models on July–September 2012, picked the one with the lowest RMSE, and only then scored it once on October–December 2012.

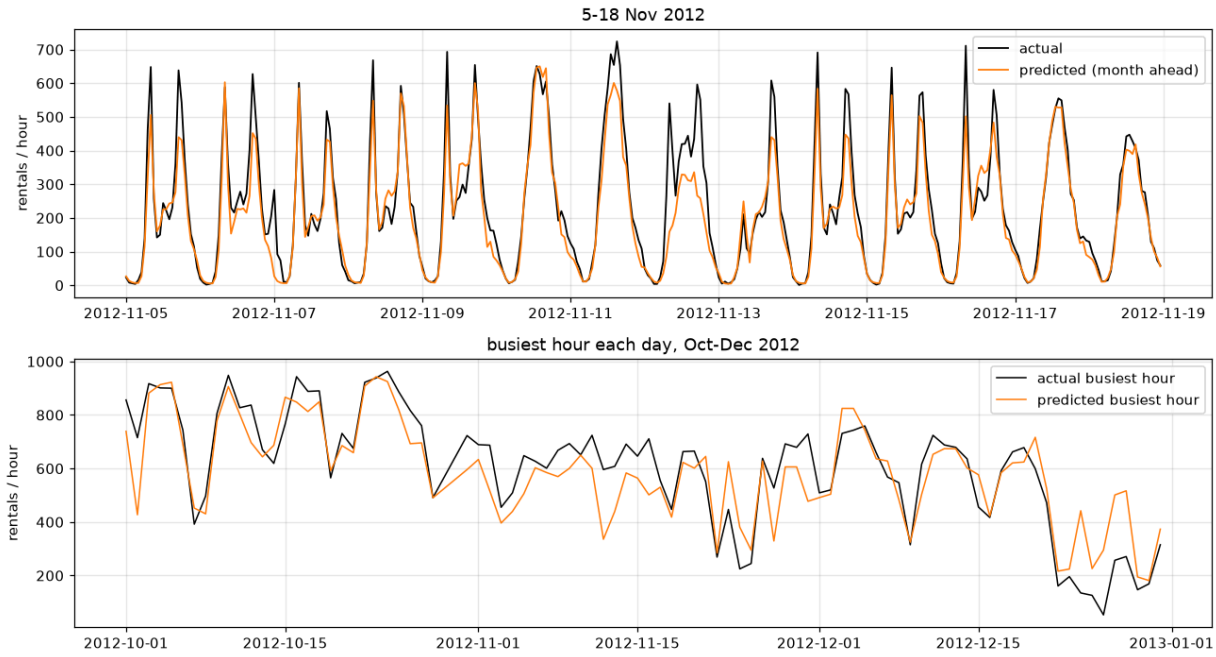
Model (Jul–Sep 2012)	MAE	RMSE	WAPE %
boosting, poisson, 15 leaves	35.0	55.9	12.1
boosting, poisson, 31 leaves	35.4	56.8	12.3
boosting, squared error, 31 leaves	36.9	57.0	12.8

boosting, squared error, 15 leaves	38.1	57.9	13.2
poisson GLM, alpha=0.01	44.7	69.6	15.5
baseline	119.7	160.8	41.5

Gradient boosting with a Poisson loss and small trees (15 leaves) won. On the held-back months it scored:

Oct–Dec 2012	MAE	RMSE	WAPE %
Chosen model	**37.9**	**61.0**	**17.2**
Baseline (average per hour and day type)	78.2	118.5	35.6

That is about half the baseline's error. Hour of day matters most, then temperature and working day. The model runs slightly low on average because demand kept growing through 2012, and it can't foresee storms. It was then retrained on all usable data and saved for the app.



Top: actual vs forecast for two weeks in November 2012. Bottom: each day's busiest hour, actual vs forecast.

Turning rides into bikes

A bike can do many rides a day, so I sized the fleet on the busiest hour. Bikes on the road = rides per hour × trip length ÷ 60 (Little's law). I then added a 25% reserve and allowed for 10% of bikes being in repair.

Busy hour used	Rides/hour	Bikes on the road	Bikes needed
95th percentile hour, 2012	663	221	**307**
99th percentile hour, 2012	835	278	**387**
Busiest hour in the data	977	326	**453**
Busiest predicted hour, test months	943	314	**437**

Changing the assumptions (trip length 10–45 minutes, reserve 0–100%) moves the busiest-hour figure between 181 and 1,629 bikes, still under 2,000. The fleet only becomes too small if trips average over 88 minutes, or if peak demand grows by roughly 340%.

D. The operations prototype

A small Streamlit app where a manager types questions in everyday English. A rule-based parser (regular expressions, no external AI) picks out dates, hours, months, weather words and day types, then answers from the saved forecasts, the trained model and the bike formula.

You type	It answers
How many bikes do we need on 2012-10-10?	about 420 bikes, peak of 906 rides at 5 pm, with the working shown
How many bikes do we need on a rainy weekday in July?	about 6,078 rides that day, peak around 681, 316 bikes
What was the busiest hour on 2012-11-07?	8 am: 601 actual, 585 predicted
Which stations run out of bikes?	explains that the data has no station information

It also has a what-if planner with sliders for date, weather, temperature and growth, and sidebar sliders for the fleet assumptions. If a question is ambiguous (like "10/11/2012" or "at 8") it asks instead of guessing.

E. Assumptions

Assumption	Used for	How solid
Average trip lasts 20 minutes	Bikes needed	Guess; tested from 10 to 45 min
Keep a 25% reserve on top of bikes in use	Bikes needed	Guess; tested from 0 to 100%
10% of bikes are in repair at any time	Bikes needed	Guess
The fleet is 2,000 bikes	Comparison	Fictional figure from the case
Missing night hours are probably zero-ride hours	Interpretation only	Likely, can't be proven
Recorded weather stands in for a weather forecast	Model testing	Real accuracy would be a bit lower
AI assistance is permitted in this round	How the work was done	Taken as given

F. What I would do next

1. Pilot forecast-led rebalancing for 4–6 weeks in the worst areas. Stock residential stations before 7 am on working days, clear full business-district stations before the evening rush, and move effort to midday at weekends. Compare complaints and empty-station minutes against similar areas.
2. Count usable bikes before each peak and cut repair times.
3. Start logging station-level bike counts, trip start and end points, and failed rental attempts.
4. Buy bikes only for stations that still run dry after the pilot. Re-check fleet size if peak demand grows sharply.

Limits to keep in mind: the data has citywide totals only and records completed rides only, so people who couldn't find a bike are invisible and real demand may be higher than shown.

A note on how this was made

I worked on the understanding that AI tools were allowed for this round, and I did use one (Claude, by Anthropic). The thinking behind the project is mine: framing the problem and my Round 1 reasoning (written before any AI use), the hypothesis, the decisions about missing data and honest testing, which assumptions to make, and the conclusions. The AI did most of the hands-on work under my direction: writing and running the code, training the models, building the app, making the charts and helping draft this report. I reviewed what it produced before using it. All the figures here come from code that was actually run, which is in the submitted notebooks.

Running the project

```
pip install -r requirements.txt
streamlit run app.py        # the operations app
# full analysis and training: notebooks/02_bike_demand_analysis.ipynb (saves the model to model/)
```